

Homework 5

1. Sort the distinct numbers 1,2,3,4,5,6,7,8 into groups.

(a) Into 3 nonempty *indistinguishable* groups

Easily, we observe this is the number of set partitions of an 8-element set into 3 nonempty unlabeled blocks $S(8, 3)$. We wish to use the inclusion-exclusion principle:

$$S(8, 3) = \frac{1}{3!} \sum_{j=0}^3 (-1)^j \binom{3}{j} (3-j)^8 = \frac{1}{6} (3^8 - 3 \cdot 2^8 + 3 \cdot 1^8 - 0^8).$$

Compute:

$$3^8 = 6561, \quad 2^8 = 256,$$

$$S(8, 3) = \frac{1}{6} (6561 - 3 \cdot 256 + 3) = \frac{1}{6} (6561 - 768 + 3) = \frac{5796}{6} = 966.$$

$$\boxed{966}$$

(b) Into 3 nonempty *labeled* groups A,B,C

We observe this counts surjections from 8 elements onto 3 labeled boxes. Labeling the 3 blocks multiplies by 3!:

$$\# = 3! S(8, 3) = 6 \cdot 966 = 5796.$$

$$\boxed{5796}$$

(c) Into at most 3 labeled groups A,B,C (empty allowed)

Trivially, we find the each of the 8 numbers independently chooses one of 3 labeled groups:

$$\# = 3^8 = 6561.$$

$$\boxed{6561}$$

2. Nine-digit sequences where 2, 5, and 8 each appear at least once

We recognize the digits are from $\{0, 1, 2, \dots, 9\}$. Let A_2 be the set of sequences missing digit 2, similarly A_5, A_8 . Utilizing the inclusion-exclusion principle:

$$\# = 10^9 - \binom{3}{1} 9^9 + \binom{3}{2} 8^9 - \binom{3}{3} 7^9.$$

$$\boxed{10^9 - 3 \cdot 9^9 + 3 \cdot 8^9 - 7^9}$$

3. Integer solutions to $A+B+C=15$ with bounds

$$0 \leq A \leq 6, \quad 0 \leq B \leq 11, \quad 2 \leq C \leq 7.$$

Let $C' = C - 2$. Then $0 \leq C' \leq 5$ and

$$A + B + C' = 13.$$

Unrestricted nonnegative solutions:

$$\binom{13+3-1}{3-1} = \binom{15}{2} = 105.$$

Now subtract violations (inclusion–exclusion on upper bounds):

$$A \geq 7: A' = A - 7 \Rightarrow A' + B + C' = 6 \Rightarrow \binom{6+2}{2} = \binom{8}{2} = 28.$$

$$B \geq 12: B' = B - 12 \Rightarrow A + B' + C' = 1 \Rightarrow \binom{1+2}{2} = \binom{3}{2} = 3.$$

$$C' \geq 6: C'' = C' - 6 \Rightarrow A + B + C'' = 7 \Rightarrow \binom{7+2}{2} = \binom{9}{2} = 36.$$

Intersections:

$$A \geq 7, B \geq 12 \Rightarrow A' + B' + C' = -6 \text{ (none),}$$

$$A \geq 7, C' \geq 6 \Rightarrow A' + B + C'' = 0 \Rightarrow 1 \text{ solution,}$$

$$B \geq 12, C' \geq 6 \Rightarrow A + B' + C'' = -5 \text{ (none).}$$

Therefore,

$$105 - (28 + 3 + 36) + 1 = 105 - 67 + 1 = 39.$$

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4. ARRANGEMENT: exactly two pairs of consecutive identical letters

We observe the word ARRANGEMENT has 11 letters with multiplicities

$$A^2, R^2, N^2, E^2, G, M, T.$$

There are 4 letter-types that could form an adjacent identical pair: A, R, N, E . We wish for exactly two of these four types to appear as a consecutive pair.

Fix which two types are consecutive (say A and R); later multiply by $\binom{4}{2}$. Treat each forced consecutive pair as a block: $[AA]$ and $[RR]$. Then we arrange the 9 objects:

$$[AA], [RR], N, N, E, E, G, M, T,$$

so

$$\#(\text{with } AA, RR \text{ consecutive}) = \frac{9!}{2!2!} = 90720.$$

Now, we wish to enforce that NN is *not* consecutive and EE is *not* consecutive via inclusion–exclusion principle.

If NN is also consecutive, use block $[NN]$:

$$[AA], [RR], [NN], E, E, G, M, T \Rightarrow \frac{8!}{2!} = 20160.$$

Similarly if EE is consecutive:

$$[AA], [RR], N, N, [EE], G, M, T \Rightarrow \frac{8!}{2!} = 20160.$$

If both NN and EE are consecutive:

$$[AA], [RR], [NN], [EE], G, M, T \Rightarrow 7! = 5040.$$

So for a fixed choice of the two consecutive types:

$$90720 - 20160 - 20160 + 5040 = 55440.$$

Choose which two of $\{A, R, N, E\}$ are consecutive:

$$\binom{4}{2} \cdot 55440 = 6 \cdot 55440 = 332640.$$

$$\boxed{332640}$$